PAPER<i>393 — SCm Reactor Efficiency with κ-Decay: E</i>react = (ρ<i>SCm·v</i>SCm<super> 2</super>/ρ_A)·exp(−κt)

Session: *107 (groksharecfdcad2f5.txt deep re-analysis pass)*

PAPER*393 — SCm Reactor Efficiency with* κ*-Decay:* Ereact = (ρ*SCm·v*SCm^2/ρ_A)·exp(−κt)

Source: grok*share*cfdcad2f5.txt, lines ~200–1400 (C++ UQFF simulation code, `compute_Ereact()`) **Section:** `CelestialBody.cpp`, `compute_Ereact()` function **Session:** 107 (grok*share*cfdcad2f5.txt deep re-analysis pass) **CP4 Class:** `SCmReactorEfficiencyDecayCalculator` (CP4 #44)

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1. Overview

The UQFF framework models the **Superconducting Matter (SCm) reactor efficiency** as a kinetic energy density ratio that decays exponentially over time. This formula was partially present in PAPER*387 (v*SCm = 0.99c update) but the **complete functional form** including the `\kappa`-decay time constant was not formalized in any existing paper.

PAPER*393 provides the complete E*react equation and demonstrates its role as the **dominant amplification factor** in Ug2, Ug3, Um, and Ug4i terms, and confirms the `t=0` value of `8.808\times10^{54}` J.

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2. The E_react Formula

2.1 Master Equation

```
\boxed{E_{\text{react}}(t) = \frac{\rho_{\text{SCm}} \cdot v_{\text{SCm}}^2}{\rho_A}
\cdot e^{-\kappa t}}
```

2.2 Parameter Definitions

Parameter	Value	Physical Meaning
`\rho_{\text{SCm}}`	`1\times10^{15}` kg/m^3	SCm density (neutron star core density)
`v_{\text{SCm}}`	`0.99c = 2.968\times10^8` m/s	SCm streaming velocity (PAPER_387)
`\rho_A`	`1\times10^{-23}` kg/m^3	Local Aether density
`\kappa`	`5\times10^{-4}` day^{-1} = `5.787\times10^{-9}` s^{-1}	SCm decay rate constant
`t`	simulation time (days)	Time since initialization

2.3 Evaluation at t = 0

```
E_{\text{react}}(0) = \frac{10^{15} \times (2.968\times10^8)^2}{10^{-23}} \cdot e^0
```

```
= \frac{10^{15} \times 8.808\times10^{16}}{10^{-23}} =
\frac{8.808\times10^{31}}{10^{-23}} = 8.808\times10^{54} \text{ J}
```

This is the **peak reactor efficiency** at simulation start.

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3. Connection to Lorentz Factor

From PAPER_387, $v_{\text{SCm}} = 0.99c$ gives Lorentz factor $\gamma = 7.089$:

$$\gamma = \frac{1}{\sqrt{1-v^2/c^2}} = \frac{1}{\sqrt{1-0.99^2}} = \frac{1}{\sqrt{0.0199}} \approx 7.089$$

The kinetic energy density in the numerator is enhanced by relativistic effects:

- Classical: $\rho_{\text{SCm}} v^2/2 = 4.404\times10^{31}$ J/m^3
- Relativistic KE: $(\gamma-1)\rho_{\text{SCm}} c^2 = 6.089 \times 1.125\times10^{47} \approx 6.85\times10^{47}$ J/m^3

The formula uses the **non-relativistic kinetic form** ρv^2 (without the 1/2 factor) as a convenient dimensionless efficiency proxy scaled against ρ_A.

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4. Time Evolution

4.1 Decay Timeline

$$E_{\text{react}}(t) = 8.808\times10^{54} \cdot e^{-5\times10^{-4} t}$$

Time (days)	E_react (J)	Fraction of peak
0	8.808×10^{54}	1.000
1000	$8.808\times10^{54} \times e^{-0.5}$ = 5.340×10^{54}	0.606
2000	$8.808\times10^{54} \times e^{-1}$ = 3.240×10^{54}	0.368
13,816	$8.808\times10^{54} \times e^{-6.908}$ = 8.808×10^{51}	0.001

The **e-folding time** is $\tau = 1/\kappa = 2000$ days ≈ 5.48 years.

4.2 Half-Life

$$t_{1/2} = \frac{\ln 2}{\kappa} = \frac{0.693}{5\times10^{-4}} \approx 1386 \text{ days} \approx 3.8 \text{ years}$$

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5. Role in UQFF Field Terms

E_react appears as a **multiplier** in four major UQFF terms:

5.1 Ug2 — Charge-Reactivity Term

$$U_{g2} = k_2 \cdot (Q_A + Q_{UA}) \cdot \frac{M_s}{r^2} \cdot S(r) \cdot w \cdot H_{\text{SCm}} \cdot E_{\text{react}}$$

5.2 Ug3 — String Rotation Term

$$U_{g3} = k_3 \cdot B_j \cdot \cos(\omega_s t\pi) \cdot P_{\text{core}} \cdot E_{\text{react}}$$

5.3 Um — Magnetic String Term

$$U_m = \frac{\mu_j}{r_j} \cdot (1-e^{-\gamma t}\cos(\pi t_n)) \cdot \Phi \cdot n_{\text{strings}} \cdot P_{\text{SCm}} \cdot E_{\text{react}}$$

5.4 Simulation Dominance

Ug3 at $t=0$ for the Sun:

$k_3 = 1.8$, $B_j \approx 10^3$ T, $P_{\text{core}} = 1.0$

$$U_{g3} = 1.8 \times 10^3 \times 1 \times 1.0 \times 8.808\times10^{54} \approx 1.6\times10^{58}$$

This matches the simulation output $U_{g3}(\text{Sun}) \sim 10^{58}$, confirming E_react as the **dominant amplitude driver** of all UQFF field terms.

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6. Physical Interpretation

6.1 Energy Density Ratio Interpretation

$$E_{\text{react}} = \frac{\rho_{\text{SCm}} v_{\text{SCm}}^2}{\rho_A}$$

This is the ratio of SCm kinetic energy density to Aether energy density. When this ratio is large (as at $t=0$: $\sim 10^{54}$), the SCm is highly energetic relative to the Aether medium, driving strong field interactions. As SCm loses energy (κ-decay), the Aether coupling weakens.

6.2 κ Physical Meaning

The rate $\kappa = 5\times10^{-4}$ day^{-1} represents the **SCm energy dissipation rate** — the timescale over which relativistic SCm slows down through interactions with the Aether field. This corresponds to PAPER_387's calibrated value for neutron star spindown physics.

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7. Comparison to Existing Papers

Paper	Formula	Distinction
PAPER_387	$v_{\text{SCm}} = 0.99c$ update	Only velocity parameter, no E_react formula
PAPER_302	Reactor energy concept	Abstract; no functional form
PAPER_393	$E_{\text{react}} = (\rho_{\text{SCm}}v^2/\rho_A)e^{-\kappa t}$	**Complete functional form + numerical value**

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8. C++ Implementation

```
double compute_Ereact(double t, double rho_SCm, double v_SCm, double rho_A, double kappa)
{
if (rho_A <= 0.0) throw std::runtime_error("Invalid rho_A value");
return (rho_SCm * v_SCm * v_SCm / rho_A) * std::exp(-kappa * t);
}
```

```
// Parameters: rho_SCm=1e15, v_SCm=2.968e8 (0.99c), rho_A=1e-23, kappa=5e-4
// E_react(0) = 1e15 * (2.968e8)² / 1e-23 = 8.808e54 J
```

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9. Summary

PAPER_393 formalizes $E_{\text{react}}(t) = (\rho_{\text{SCm}} v_{\text{SCm}}^2 / \rho_A) \cdot e^{-\kappa t}$ with peak value 8.808×10^{54} J at $t=0$ and e-folding decay time $\tau = 2000$ days. This function is the dominant amplitude multiplier in Ug2, Ug3, Um, and Ug4i UQFF terms, confirmed by simulation outputs showing $F_U(\text{Sun}) \approx -2.064\times10^{59}$ driven primarily by $U_{g3} \sim 10^{58}$.
